

Lecture 1 2nd quantisation for electrons

1. Expansion of a many-electron ψ

- $\{\psi_n(x), n=1,2,\dots\} \leftarrow$ set of functions

$$\sum_n \psi_n(x) \psi_n^*(x') = \delta(x-x') \leftarrow \text{complete} \quad (1)$$

$$\int \psi_n^*(x) \psi_{n'}(x) dx = \langle \psi_n | \psi_{n'} \rangle = \delta_{nn'} \leftarrow \text{orthonormal} \quad (2)$$

Then

$$\psi(x_1, \dots, x_N) = \sum_{n_1, n_2, \dots} c_{n_1, n_2, \dots} \psi_{n_1}(x_1) \psi_{n_2}(x_2) \dots \quad (3)$$

$$c_{n_1, n_2, \dots} = \int \psi(x_1, x_2, \dots) \psi_{n_1}^*(x_1) \psi_{n_2}^*(x_2) \dots dx_1 dx_2 \dots \quad (4)$$

- Consider two sets of indices

$$n = (n_1, n_2, \dots) \text{ and } m = (m_1, m_2, \dots)$$

containing the same indices, but in different order:

$$m = \hat{P} n$$

$$\uparrow \text{permutation } \begin{pmatrix} n_1 & n_2 & \dots \\ m_1 & m_2 & \dots \end{pmatrix}$$

Then:

$$c_{m_1, m_2, \dots} = \int \psi(x_1, x_2, \dots) \psi_{n_1}^*(y_1) \psi_{n_2}^*(y_2) \dots dx_1 dx_2, \dots$$

$$(y_1, y_2, \dots) = \hat{P}^{-1} (x_1, x_2, \dots)$$

$$\hookrightarrow c_{m_1, m_2, \dots} = \int \psi(x_1, x_2, \dots) \hat{P}_x^{-1} [\psi_{n_1}^*(x_1) \psi_{n_2}^*(x_2) \dots] dx_1 dx_2, \dots$$

$\uparrow$ acts on arguments $(x_1, x_2, \dots)$

Example $n = (1, 2, 3)$, $m = (2, 3, 1) = \hat{P} n$, $\hat{P} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ \uparrow & \uparrow & \uparrow \\ 2 & 3 & 1 \end{pmatrix}$

$$c_{123} = \int \psi(x_1, x_2, x_3) \psi_1^*(x_1) \psi_2^*(x_2) \psi_3^*(x_3) dx_1 dx_2 dx_3$$

$$C_{231} = \int \psi(\dots) \psi_2^*(x_1) \psi_3^*(x_2) \psi_1^*(x_3) dx_1 dx_2 dx_3$$

$$= \int \psi(\dots) \hat{P}_x^{-1} [\psi_1^*(x_1) \psi_2^*(x_2) \psi_3^*(x_3)] dx_1 dx_2 dx_3$$

with $\hat{P}_x^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = (132)$, where

$$(132)(123) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = \text{no permutation}$$

↑
times

Hence,

$$C_{n_1 n_2 \dots} = \int (\hat{P}_x \psi(\dots)) \psi_{n_1}^*(x_1) \psi_{n_2}^*(x_2) \dots dx_1 dx_2 \dots$$

$$= \epsilon_P \int \psi(\dots) \psi_{n_1}^*(x_1) \psi_{n_2}^*(x_2) \dots dx_1 dx_2 \dots$$

$$= \epsilon_P C_{n_1 n_2 \dots}, \quad \epsilon_P = (-1)^P \leftarrow \text{parity of } \hat{P} \quad (5)$$

Collecting all terms in the summation having the same set of indices.

$$\sum_{\text{set}(n_1, n_2, \dots)} C_{n_1 n_2 \dots} \psi_{n_1}^*(x_1) \psi_{n_2}^*(x_2) \dots$$

No! terms

$$= C_{\underbrace{\text{set}(n_1 n_2 \dots)}_{\text{some chosen order of indices}}} \underbrace{\sum_{\hat{P} \in S_N} \epsilon_P \hat{P} \psi_{n_1}^*(x_1) \psi_{n_2}^*(x_2) \dots}_{\text{Slater Determinant}}$$

$$\psi = \sum_{\text{sets}} C_{\text{sets}} \Phi_{\text{set}}(x_1, \dots, x_N) \leftarrow \text{the sum over "configurations"} \quad (6)$$

$$\Phi_K(x_1, \dots, x_N) = \frac{1}{\sqrt{N!}} \sum_P \epsilon_P \hat{P} [\psi_{n_1}(x_1) \dots \psi_{n_N}(x_N)] \quad (7)$$

$K = (n_1, n_2, \dots)$

(2)

$$\hat{A} \Phi_k(x_1, x_2, \dots) = \Phi_k, \quad \hat{P} \Phi_k = \varepsilon_p \Phi_k$$

$$\begin{aligned} \langle \hat{p}_x \Phi_k | \Phi_{k'} \rangle &= \int \hat{p}_x [\Phi_k^* \hat{p}_x^{-1} \Phi_{k'}] dx_1 \dots \\ &= \int \Phi_k^* (\hat{p}_x^{-1} \Phi_{k'}) \underbrace{dx_1 \dots}_{dX} = \langle \Phi_k | \hat{p}_x^{-1} \Phi_{k'} \rangle \quad (9) \end{aligned}$$

$$\langle \underbrace{\Phi_k}_{\text{}} | \hat{O} | \Phi_{k'} \rangle = \frac{1}{\sqrt{N!}} \sum_p \varepsilon_p \langle \hat{p}_x \tilde{\Phi}_k | \hat{O} | \Phi_{k'} \rangle$$

$$\sqrt{N!} \hat{A} \underbrace{\tilde{\Phi}_k}_{\text{product of orbitals only}} = \frac{1}{\sqrt{N!}} \sum_p \varepsilon_p \langle \tilde{\Phi}_k | \hat{P}_x^{-1} | \Phi_{k'} \rangle$$

$$\equiv \langle \Phi_k | \hat{O} | \Phi_{k'} \rangle \sqrt{N!} \quad (10)$$

$$(\hat{P}_X^{-1} \Phi_{K'} = \varepsilon_{P^{-1}} \Phi_{K'} \equiv \varepsilon_P \Phi_{K'} \text{ and } \sum_P \varepsilon_P^2 = \sum_P 1 = N!))$$

- One-particle operator

$$\hat{O}_1 = \sum_{i=1}^N \hat{h}(x_i) \leftarrow \left(\text{symmetrical for any } \hat{P} \right) \quad (10)$$

$$\langle \Phi_k | \hat{O}_1 | \Phi_k \rangle = \sqrt{N!} \sum_{i=1}^N \langle \tilde{\Phi}_k | \hat{h}(x_i) | \tilde{\Phi}_k \rangle$$

$$= \sum_i \sum_p \epsilon_p \langle \hat{\Phi}_k | \hat{h}(x_i) | \hat{p} \hat{\Phi}_{k'} \rangle$$

- ↳ acts on indices in the product

Let $k = (n_1 n_2 \dots)$, $k' = (m_1 m_2 \dots)$, then $\hat{P} k' = (m_1' m_2' \dots)$

$$\hookrightarrow \langle \Phi_k | \hat{O}_1 | \Phi_{k'} \rangle = \sum_i \sum_{p_i} \epsilon_{p_i} \underbrace{\langle \psi_{n_1} | \psi_{m_1'} \rangle}_{\delta_{n_1 m_1'}} \dots \langle \psi_{n_i} | \hat{h} | \psi_{m_i'} \rangle \dots \underbrace{\langle \psi_{n_N} | \psi_{m_N'} \rangle}_{\delta_{n_N m_N'}}$$

If $k = k'$ (diagonal): $m_1' = n_1, m_2' = n_2, \dots$, i.e. $\hat{P} k = k \rightarrow \underline{\underline{\hat{P} = \hat{E}}}$ only,

$$\boxed{\langle \Phi_k | \hat{O}_1 | \Phi_k \rangle = \sum_i \langle \psi_{n_i} | \hat{h} | \psi_{n_i} \rangle} \quad (12)$$

If $k \neq k'$ (off-diagonal): only one orbital may be different.

$k = (n_1 n_2 \dots n)$, $k' = (n_1 n_2 \dots m)$, $n \neq m$

$\hookrightarrow \hat{P} = \hat{E}$ and $i = N$

$$\hookrightarrow \boxed{\langle \Phi_k | \hat{O}_1 | \Phi_{k'} \rangle = \langle \psi_n | \hat{h} | \psi_m \rangle} \quad (13)$$

And zero in other cases!

• Slater determinants form a complete orthonormal basis

set: $\hat{h} \equiv \hat{1} = \sum_i \frac{1}{N} \rightarrow \begin{cases} \langle \Phi_k | \Phi_k \rangle = 1 \\ \langle \Phi_k | \Phi_{k'} \rangle = 0 \text{ if } k \neq k' \end{cases} \quad (14)$

~~Two-particle operators~~

~~$$\hat{Q}_2(x_1, \dots, x_N) = \frac{1}{2} \sum_{i \neq j} v(x_i, x_j) \leftarrow \text{(also symmetric)}$$~~

• Slater determinants also form a complete set:

~~$$\sum_k \Phi_k^*(x) \Phi_k(x') = \sum_{\substack{n_1 n_2 \dots \\ \text{all different}}} \Phi_k^*(x) \Phi_k(x') = \sum_{\substack{n_1 n_2 \dots \\ \text{all possible}}} \Phi_k^*(x) \Phi_k(x')$$~~

in a sense that we started from the complete expansion

• Two-particle operators

$$\hat{O}_2(X) = \frac{1}{2} \sum_{i \neq j} v(x_i, x_j) \leftarrow \text{also symmetric} \quad (15)$$

$\hookrightarrow (x_1, x_2, \dots)$

$$\langle \Phi_k | \hat{O}_2 | \Phi_{k'} \rangle = \frac{1}{2} \sum_{i \neq j} \sum_P \varepsilon_P \delta_{n_1 m_1'} \delta_{n_2 m_2'} \dots (n_i n_j | m_i' m_j') \dots \delta_{n_N m_N} \quad (16)$$

$$m' = \hat{P} m, \quad k' = (m_1 m_2 \dots), \quad k = (n_1 n_2 \dots)$$

It is zero, if more than 2 orbitals are different in k and k' .

(i) $\underline{k = k'} \rightarrow \hat{P} = \hat{E}$ or (ij) : Coulomb Exchange

$$\langle \Phi_k | \hat{O}_2 | \Phi_k \rangle = \frac{1}{2} \sum_{i \neq j} \left[(n_i n_j | n_j n_i) - (n_i n_j | n_i n_j) \right] \quad (17)$$

(ii) $\underline{k \neq k'}$, differ by one orbital only: $k = (n_1 n_2 \dots n_p n)$
 $k' = (n_1 n_2 \dots n_p m), \quad m \neq n$

$\hookrightarrow \hat{P} = \hat{E}$ or (ij) , and i or j must be N

$$\langle \Phi_k | \hat{O}_2 | \Phi_{k'} \rangle = \left(\frac{1}{2} \right) \sum_{i \neq N} \left[(n_i \underbrace{n_N}_n | \underbrace{m_N}_m n_i) - (n_i \underbrace{n_N}_n | \underbrace{n_i}_m \underbrace{m_N}_m) \right] \quad (18)$$

since both cases $i=N$ and $j=N$ give identical contribution (due to symmetry of the QC integral over permutations)

(iii) $\underline{k \neq k'}$, differ by two orbitals

$$k = (n_1 n_2 \dots n_p n \ell), \quad k' = (n_1 n_2 \dots n_p m k)$$

then i, j should be either $N-1$ or N (again, two cases)

$$\langle \Phi_k | \hat{O}_2 | \Phi_{k'} \rangle = \left(\underbrace{n_{N-1}}_n \underbrace{n_N}_\ell | \underbrace{m_N}_k \underbrace{m_{N-1}}_m \right) - \left(\underbrace{n_{N-1}}_n \underbrace{n_N}_\ell | \underbrace{m_{N-1}}_m \underbrace{m_N}_k \right) \quad (19)$$

$|0\rangle = |vac\rangle = |000\dots\rangle \leftarrow \text{nothing is occupied}$

$$|n_2\rangle = \phi_{n_1}(x_1), \quad |n_1 n_2\rangle = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_{n_1}(x_1) & \psi_{n_2}(x_1) \\ \psi_{n_1}(x_2) & \psi_{n_2}(x_2) \end{vmatrix}, \text{ etc.}$$

- Creation: N -state $\Rightarrow (N+1)$ -state via:
- $$\underbrace{a_i^\dagger |n_1 n_2 \dots n_p\rangle}_{N \text{ of } e^-} = \begin{cases} |n_1 n_2 \dots n_p i\rangle, & i \notin (n_1 n_2 \dots n_p) \\ 0, & i \in (n_1 n_2 \dots n_p) \end{cases}$$
- (21)

• If $n_1 < n_2 < n_3$: $a_{n_2}^\dagger |n_1 n_3\rangle = |n_1 n_3 n_2\rangle = -|n_1 n_2 n_3\rangle$

• $a_{n_1}^\dagger |0\rangle = |n_1\rangle$, $a_{n_2}^\dagger a_{n_1}^\dagger |0\rangle = |n_1 n_2\rangle$, etc.

- Annihilation N -state $\Rightarrow (N-1)$ -state via:

$$\left\{ \begin{aligned} a_i |n_1 n_2 \dots i \dots n_p\rangle &= \underbrace{(-1)^r}_{\substack{\rightarrow \text{to bring } i \text{ to the end of the list}}} |n_1 n_2 \dots n_p\rangle \\ a_i |n_1 n_2 \dots n_p\rangle &= 0 \quad \text{if } i \notin \text{to the list of orbitals} \end{aligned} \right. \quad (22)$$

- If $n_1 < n_2 < n_3$, then

$$a_{n_2} |u_1 u_2 u_3\rangle = -a_{n_2} |u_1 u_3 u_2\rangle = -|u_1 u_3\rangle$$

- $a_n |0\rangle = 0$

- Anticommutation relations

(i) a_i, a_j

$a_i a_i = a_i^2 = 0$ as the same state cannot be destroyed twice

$$a_i a_j | \dots i \dots j \dots \rangle = \pm a_i a_j | \dots \bar{i} \bar{j} \dots \rangle = \pm | \dots \rangle$$

while

$$a_j a_i | \dots i \dots j \dots \rangle = \pm a_j a_i | \dots ij \dots \rangle = \mp a_j a_i | \dots ji \dots \rangle = \pm | \dots \rangle$$

$$\hookrightarrow \boxed{a_i a_j + a_j a_i = 0} \quad (\text{in particular, } a_i^2 = 0) \quad (23)$$

If one (i or j) is not present in the set, the result = 0, and the operator identity still correct.

(ii) $a_i^\dagger, a_j^\dagger$ $(a_i^\dagger)^2 = a_i^\dagger a_i^\dagger = 0$ (cannot create twice)

$$a_i^\dagger a_j^\dagger |\dots i \dots j \dots\rangle = 0 \quad \text{if at least one of the states is already present}$$

$$a_i^\dagger a_j^\dagger |\dots\rangle = |\dots j i\rangle, \text{ but } a_j^\dagger a_i^\dagger |\dots\rangle = |\dots i j\rangle = -|\dots j i\rangle$$

$$\hookrightarrow \boxed{a_i^\dagger a_j^\dagger + a_j^\dagger a_i^\dagger = 0} \quad (\text{in particular, } (a_i^\dagger)^2 = 0) \quad (24)$$

(iii) $a_i, a_j^\dagger$

$$a_i a_j^\dagger |\dots\rangle = 0 \quad \text{if either } i \text{ not present, or } j \text{ is present}$$

$$a_i a_j^\dagger |\dots i \dots\rangle = a_i |\dots i \dots j\rangle = a_i (\pm |\dots j i\rangle) = \pm |\dots j\rangle$$

but $a_j^\dagger a_i |\dots i \dots\rangle = a_j^\dagger a_i (\mp |\dots i\rangle) = \mp a_j^\dagger |\dots i\rangle = \mp |\dots j\rangle$
 $\uparrow$ (one less permutation necessary)

$$\therefore a_i a_j^\dagger + a_j^\dagger a_i = 0 \quad \text{if } i \neq j$$

If $i = j$, then:

$$a_i^\dagger a_i |\dots i \dots\rangle = a_i^\dagger a_i (\pm |\dots i\rangle) = a_i^\dagger (\pm |\dots\rangle) = \pm |\dots i\rangle$$

but $\equiv |\dots i \dots\rangle$ (the same # of permutations)

$$\text{but } a_i a_i^\dagger |\dots i \dots\rangle = 0 \quad (25)$$

$$\hookrightarrow (a_i^\dagger a_i + a_i a_i^\dagger) |\dots i \dots\rangle = |\dots i \dots\rangle \rightarrow \boxed{a_i^\dagger a_i + a_i a_i^\dagger = \delta_{ij}}$$

If ψ_i is not in the list, then $a_i^\dagger a_i |\dots\rangle = 0$, but $a_i a_i^\dagger |\dots\rangle = |\dots\rangle$, still the same operator identity.

4. Operators in 2nd quantisation

$$\Psi = \sum_{\substack{n_1, n_2, \dots \\ \text{all different}}} C_{n_1 n_2 \dots} |n_1 n_2 \dots\rangle \text{ in the Fock space}$$

exact valid for any # of e^-
convenient for many e^- systems

$\uparrow \downarrow$

Slater representation

Therefore, $\hat{H}$ should be presented via $a, a^\dagger$ so that to satisfy the Slater rules.

• One-particle operators

$$\hat{O}_1(X) = \sum_{i=1}^N h(x_i) \iff \boxed{\hat{O}_1 = \sum_{nm} h_{nm} a_n^\dagger a_m} \quad (26)$$

$\langle n | \hat{h} | m \rangle$

an infinite summation

Consider

$$\begin{aligned} \langle n_1 \dots n_p | \hat{O}_1 | n'_1 \dots n'_p \rangle &= \sum_{nm} h_{nm} \langle n_1 \dots n_p | a_n^\dagger a_m | n'_1 \dots n'_p \rangle \\ &= \sum_{nm} h_{nm} \langle \chi_n | \chi'_m \rangle \end{aligned}$$

$$\chi'_m = a_m |n'_1 \dots n'_p\rangle = a_m a_{n'_1}^\dagger \dots a_{n'_p}^\dagger |0\rangle$$

$$\chi_n = a_n |n_1 \dots n_p\rangle = a_n a_{n_1}^\dagger \dots a_{n_p}^\dagger |0\rangle$$

$\langle \chi_n | \chi'_m \rangle \neq 0$ if both sets/lists are identical. $\rightarrow$ (original list do not differ by more than 1 orbital)
 $\chi_n \neq 0$ and $\chi'_m \neq 0$ if n, m belong to the respective lists.

(i) diagonal matrix element

$$(n_1, \dots, n_p) = (n'_1, \dots, n'_p) \Rightarrow n=m \Rightarrow \sum_n h_{nn} = \sum_{n \in (n_1, \dots, n_p)} \langle n | \hat{h} | n \rangle$$

(ii) one orbital is different

$(n_1, \dots, n_p, l)$ and $(n_1, \dots, n_p, k)$, $k \neq l$ (at the end of the lists)

$$\hookrightarrow \chi_n = \chi'_m \text{ if } n=l \text{ and } m=k \Rightarrow h_{ek} = \langle l | \hat{h} | k \rangle$$

(iii) more than 1 orbital difference $\Rightarrow 0$. $\blacktriangleleft$ (Proven)

• Two-particle operators

QC integrals

$$\hat{O}_2(X) = \frac{1}{2} \sum_{i \neq j} v(x_i, x_j) \longleftrightarrow \boxed{\hat{O}_2 = \frac{1}{2} \sum_{nm\ell k} (nm|\ell k) a_n^\dagger a_m^\dagger a_\ell a_k} \quad (27)$$

infinite

$$\langle n_1 \dots n_p | \hat{O}_2 | n'_1 \dots n'_p \rangle = \frac{1}{2} \sum_{nm\ell k} (nm|\ell k) \langle X_{mn} | X'_{\ell k} \rangle$$

$$X_{mn} = a_m a_n |n_1 \dots n_p\rangle, \quad X'_{\ell k} = a_\ell a_k |n'_1 \dots n'_p\rangle$$

$\hookrightarrow m, n \in (n_1, \dots, n_p)$, where $\ell, k \in (n'_1, \dots, n'_p)$

Also, $\langle X_{mn} | X'_{\ell k} \rangle \neq 0$ if only the two states are identical, which means the two original states differ by no more than 2 orbitals.

(i) diagonal

the same in both

$$X_{mn} = a_m a_n |n_1 \dots n_p\rangle, \quad X'_{\ell k} = a_\ell a_k |n_1 \dots n_p\rangle$$

and are identical if $m=\ell, n=k$ or $m=k, n=\ell$:

$$\hookrightarrow \frac{1}{2} \sum_{nm\ell k} \dots \Rightarrow \frac{1}{2} \sum_{n, m \in (n_1 \dots n_p)} (nm|nm) \text{ if } m=\ell, n=k$$

$$\frac{1}{2} \sum_{nm\ell k} \dots \Rightarrow -\frac{1}{2} \sum_{n, m \in (n_1 \dots n_p)} (nm|nm) \text{ if } m=k, n=\ell$$

$$(\text{since } X'_{\ell k} = X'_{nm} = -X'_{mn} \equiv -X_{mn})$$

Combining both $\rightarrow$ Slater rules.

(ii) differ by one orbital: $|n_1 \dots n_p d\rangle \neq |n_1 \dots n_p f\rangle, d \neq f$

$$X_{mn} = a_m a_n |n_1 \dots n_p d\rangle, \quad X'_{\ell k} = a_\ell a_k |n_1 \dots n_p f\rangle$$

are identical if we destroy both d and f . Four cases:

$$(a) \underline{n=d, k=f \text{ and } m=\ell} \quad |m=d, \ell=f \text{ and } n=k \quad (c)$$

$$(b) \underline{n=d, \ell=f \text{ and } m=k} \quad |m=d, k=f \text{ and } n=\ell \quad (d)$$

$$(a): X_{mn} = a_e a_d |n_1 \dots n_p d\rangle, \quad X'_{\ell k} = a_e a_f |n_1 \dots n_p f\rangle$$

$a_e |n_1 \dots n_p\rangle \qquad a_e |n_1 \dots n_p\rangle$

$$\hookrightarrow \frac{1}{2} \sum_{d \in (n_1 \dots n_p)} (d|e|f)$$

$$(b): X_{mn} = \alpha_k \alpha_d |n_1 \dots n_p d\rangle = \alpha_k |n_1 \dots n_p\rangle$$

$$X'_{ek} = \alpha_f \alpha_k |n_1 \dots n_p f\rangle = -\alpha_k |n_1 \dots n_p\rangle$$

$$\hookrightarrow -\frac{1}{2} \sum_k (dk|fk)$$

The same result in other 2 cases (c), (d). $\rightarrow$ all 4 give the required Slater rules.

(iii) ~~more than~~ 2 orbitals different ~~$\neq 0$~~ . different

$$X_{mn} = \alpha_m \alpha_n |n_1 \dots n_p df\rangle, X'_{ek} = \alpha_p \alpha_k |n_1 \dots n_p \overbrace{d'f'}^{\text{different}}\rangle$$

then $(m,n) = (df)$ or (fd) , and $(lk) = (d'f')$ or $(f'd')$

$\hookrightarrow$ Slater rules again (4 cases to consider as above). $\blacktriangleleft$

5. Density operator

$$\hat{O}_1^{(x)}(X) = \sum_{i=1}^N \delta(x-x_i)$$

Indeed,

$$\langle \Psi | \hat{O}_1^{(x)} | \Psi \rangle = \sum_{i=1}^N \int \Psi^*(X) \delta(x-x_i) \Psi(X) dX$$

$$= \sum_{i=1}^N \int \Psi^*(x_1, \dots, x_{i-1}, x, x_{i+1}, \dots) \Psi(x_1, x_{i-1}, x, x_{i+1}, \dots) \underbrace{dx_1 \dots dx_i}_{\text{without } dx_i}$$

$$\boxed{\int \rho(x) dx = N}$$

$$\frac{1}{N} \rho(x)$$

$\hookrightarrow$ diagonal reduced density matrix;
does not depend on i used

$$\equiv N \frac{1}{N} \rho(x) = \rho(x) \leftarrow \begin{pmatrix} \text{a probability to find an } e^- \\ \text{at } x \end{pmatrix}$$

Hence,

$$\hat{\rho}(x) = \sum_{i=1}^N \delta(x-x_i) \rightarrow \hat{\rho}(x) = \sum_{n,m} \langle n | \delta(x-x_1) | m \rangle a_n^\dagger a_m$$

↳

$$\boxed{\hat{\rho}(x) = \sum_{n,m} \psi_n^*(x) \psi_m(x) a_n^\dagger a_m} \equiv \psi^\dagger(x) \psi(x) \quad (28)$$

where

$$\boxed{\psi(x) = \sum_n \psi_n(x) a_n} \leftarrow \text{(the field operator)} \quad (29)$$

6. Total energy via field operators

$$E = \langle \Psi | \hat{H} | \Psi \rangle = E_1 + E_2$$

$$E_1 = \sum_{n,m} h_{nm} \langle a_n^\dagger a_m \rangle = \int dx \psi^\dagger(x) \hat{h}(x) \psi(x) \quad (30)$$

$$\begin{aligned} E_2 &= \frac{1}{2} \sum_{nmkl} (nm|kl) a_n^\dagger a_m^\dagger a_k a_l \\ &= \frac{1}{2} \int dx dx' \psi^\dagger(x) \psi^\dagger(x') v(x, x') \psi(x') \psi(x) \end{aligned} \quad (31)$$

7. Current operator

We consider electrons ~~without~~ subjected to external fields

$$\star \quad \hat{j}(x) = -\frac{i\hbar}{2m} \left[\psi^\dagger(x) \nabla \psi(x) - (\nabla \psi^\dagger(x)) \psi(x) \right] \quad (32)$$

No proof. A general one is a bit lengthy, but we shall not use this expression anyway.